NexTurn
Realize Your True North

CAPSTONE PROJECT

Evently

Campus event management: discovery, booking, approval and check-in in one role-based platform

TEAM MEMBERS

1. Mutyala Satya Aditya
2. Chanikya Nelli
3. Eslavath Rajesh

UNDER THE GUIDANCE OF

Rishabh Verma

01 / PROBLEM

Campus events are scattered

The problem

- Event details, capacity, approvals and check-in are handled separately
- Full events leave students with no queue
- Organizers lack live attendance figures

Objectives

- One place to browse approved events
- Role-based access for students, organizers, admins
- No overbooking or duplicate bookings
- Waitlist with automatic promotion
- Event-level metrics for organizers

02 / USERS

Three roles, focused workflows

Student

- Register and log in
- Search and filter approved events
- Book a seat or join the waitlist
- View tickets and cancel

Organizer

- Create and edit event drafts
- Submit events for review
- View dashboard statistics
- Check in tickets for own events

Administrator

- Approve or reject events
- Manage venues
- List and create users
- Change user roles

03 / WORKFLOW

From draft to attendance

Draft → Pending → Approved → Booked → Checked in

Event rules

- Organizer submits a draft; admin approves or rejects with a note
- Rejected events can be edited and resubmitted
- Capacity must be positive and within venue capacity

Booking rules

- Seat booked if capacity allows, else waitlisted
- Cancellation releases the seat and promotes the first in queue
- Duplicate tickets are blocked; used tickets cannot be rescanned

04 / ARCHITECTURE

Separated frontend and API

Presentation	React 18, Vite, React Router, QR ticket UI
API	Flask REST with Blueprints, JWT access and refresh tokens, role checks
Persistence	MongoDB: users, venues, events, tickets, waitlist, check-ins, token blocklist
Packaging	Dockerfiles and Docker Compose for repeatable setup

Collections and route families

Data model

- users, venues, events
- tickets, waitlist, check-ins
- token_blocklist for revoked JWTs
- Unique indexes on emails, venue names, ticket codes

API families

- /api/auth: register, login, refresh, logout
- /api/student: events, book, tickets, waitlist
- /api/organizer: events, dashboard, check-in
- /api/admin: approvals, venues, users, roles
- /api/health: database ping

Risks, testing and next steps

Items to verify

- Secrets kept out of source control
- UTC-consistent date handling
- Waitlist and seat-count concurrency
- Venue deletion safety
- No automated tests yet

Next steps

- OpenAPI specification
- Unit and integration tests
- Email notifications
- Pagination and audit history
- Production hardening: HTTPS, rate limiting